

BIS514 – System Analysis, Modeling, and Design

Remote Customer Onboarding System

PHASE 1 - April 2026

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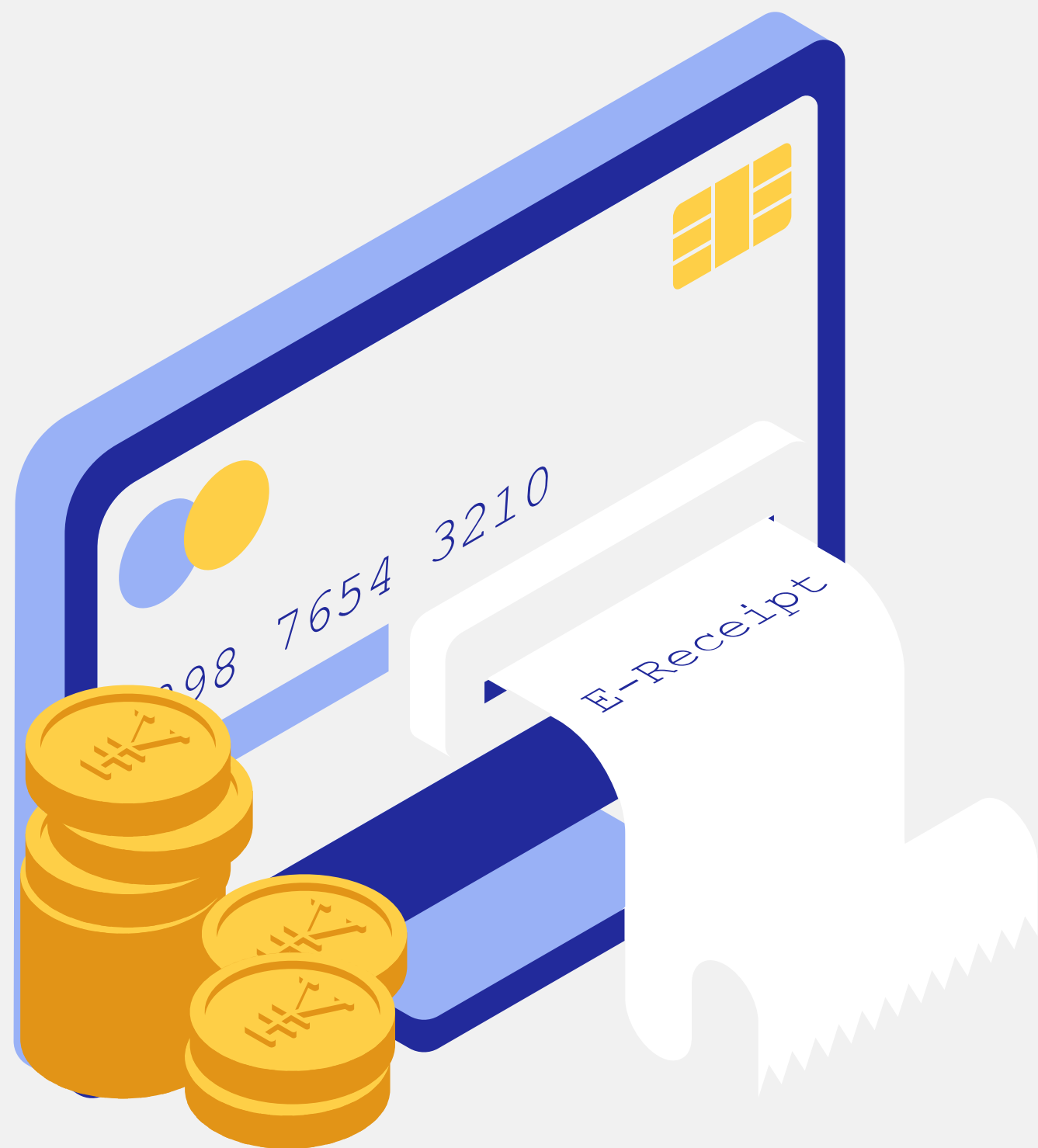


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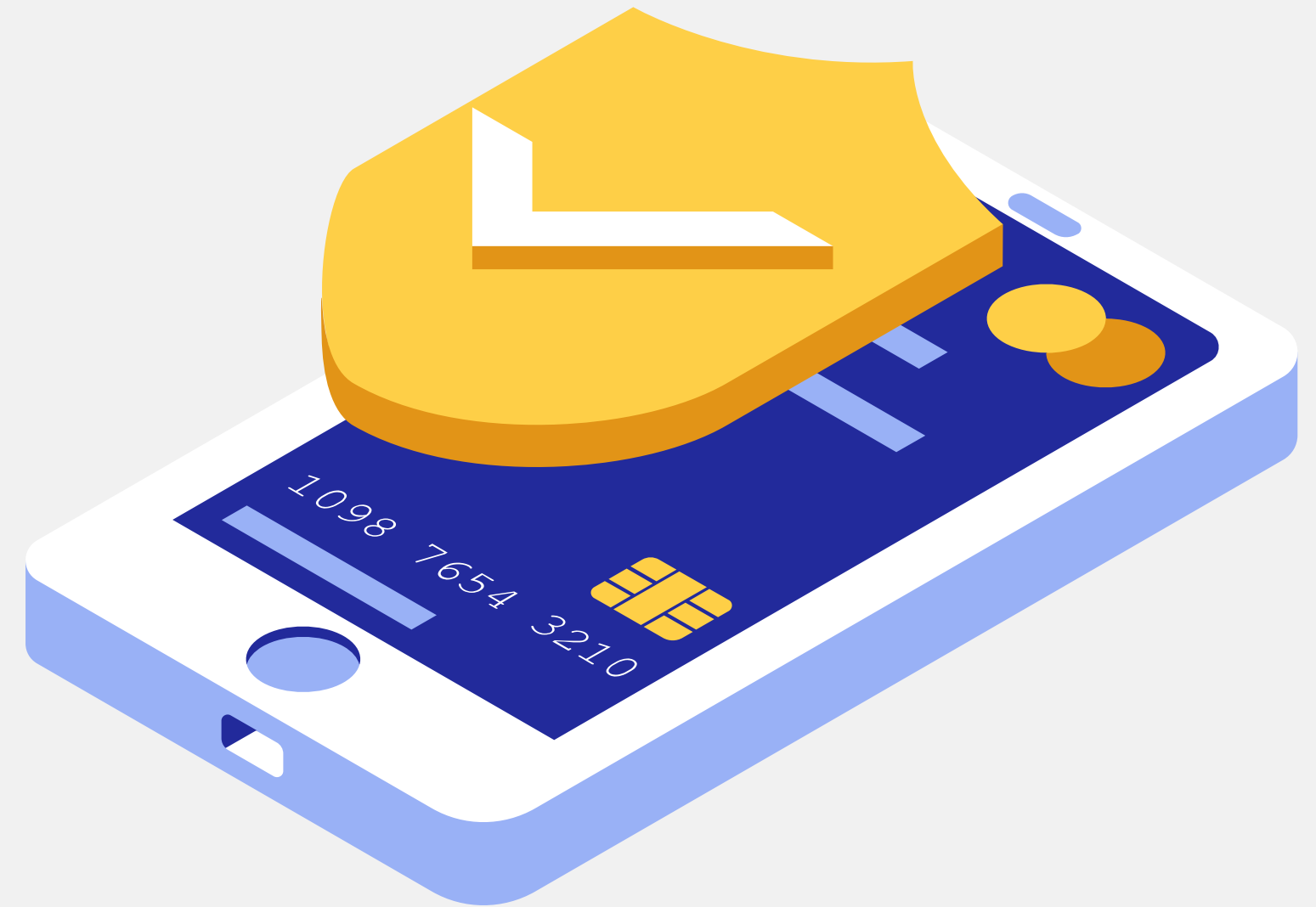
Project Overview

What is the Digital Onboarding System?

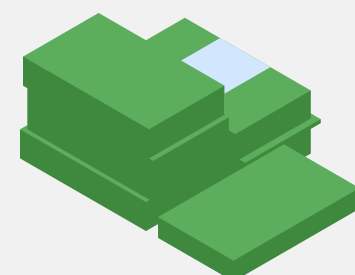
An online customer acquisition platform enabling individuals to become bank customers remotely without visiting a physical branch.

Delivered via mobile or web channels, it provides a fast, secure, and user-friendly registration process.

The main purpose of the system is to digitalize the customer onboarding journey, reduce manual workload, shorten processing time, improve customer experience, and ensure compliance with legal and banking regulations.



Requirements Collection

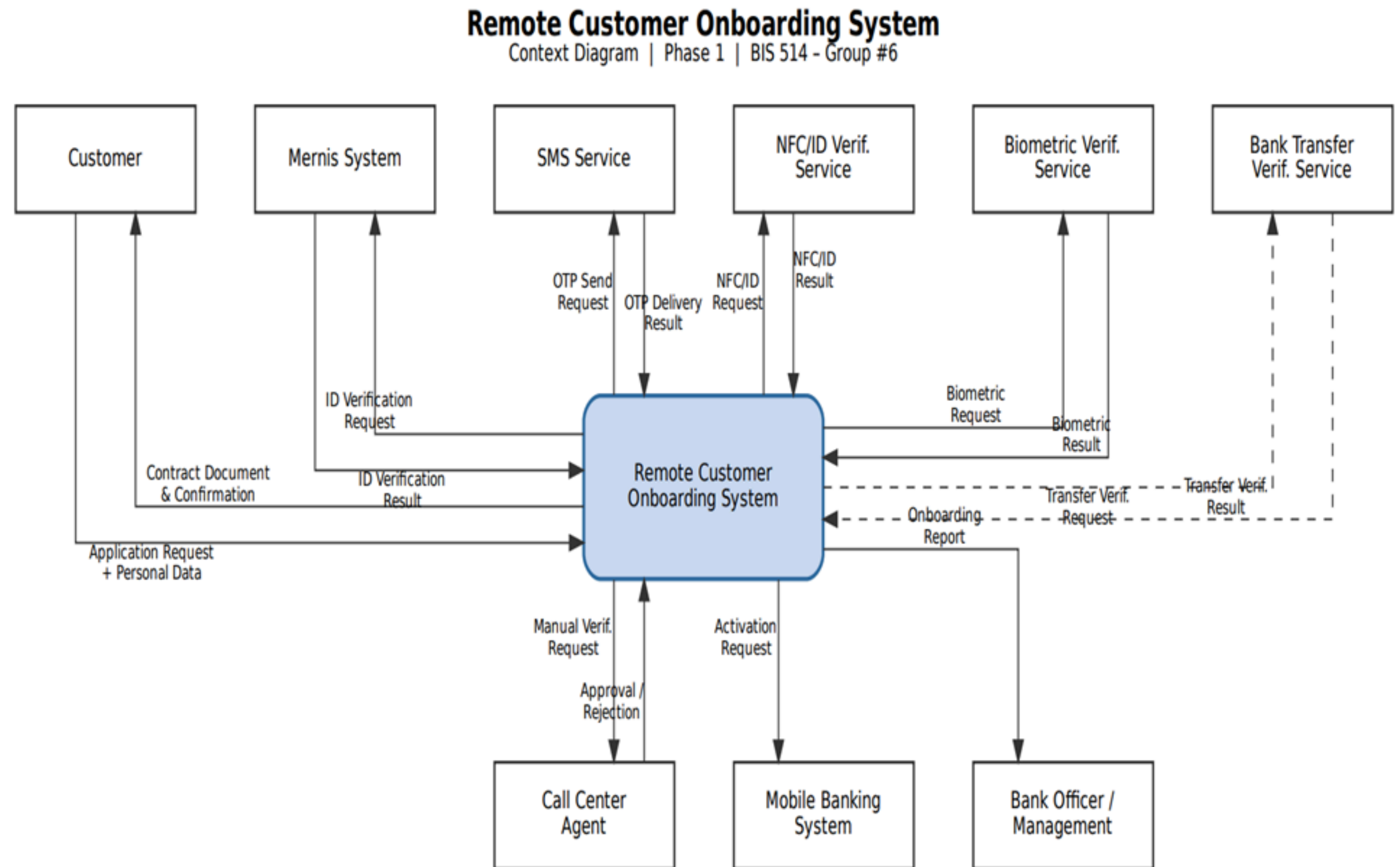


End-to-End Process Analysis	Stakeholder Identification	Data Analysis	Exception Scenarios
Workflow analyzed from application start to account activation	Customer, bank employees, call center, external services	Inputs, outputs, and data stores throughout the journey	Failed SMS, NFC errors, selfie mismatch, incomplete data

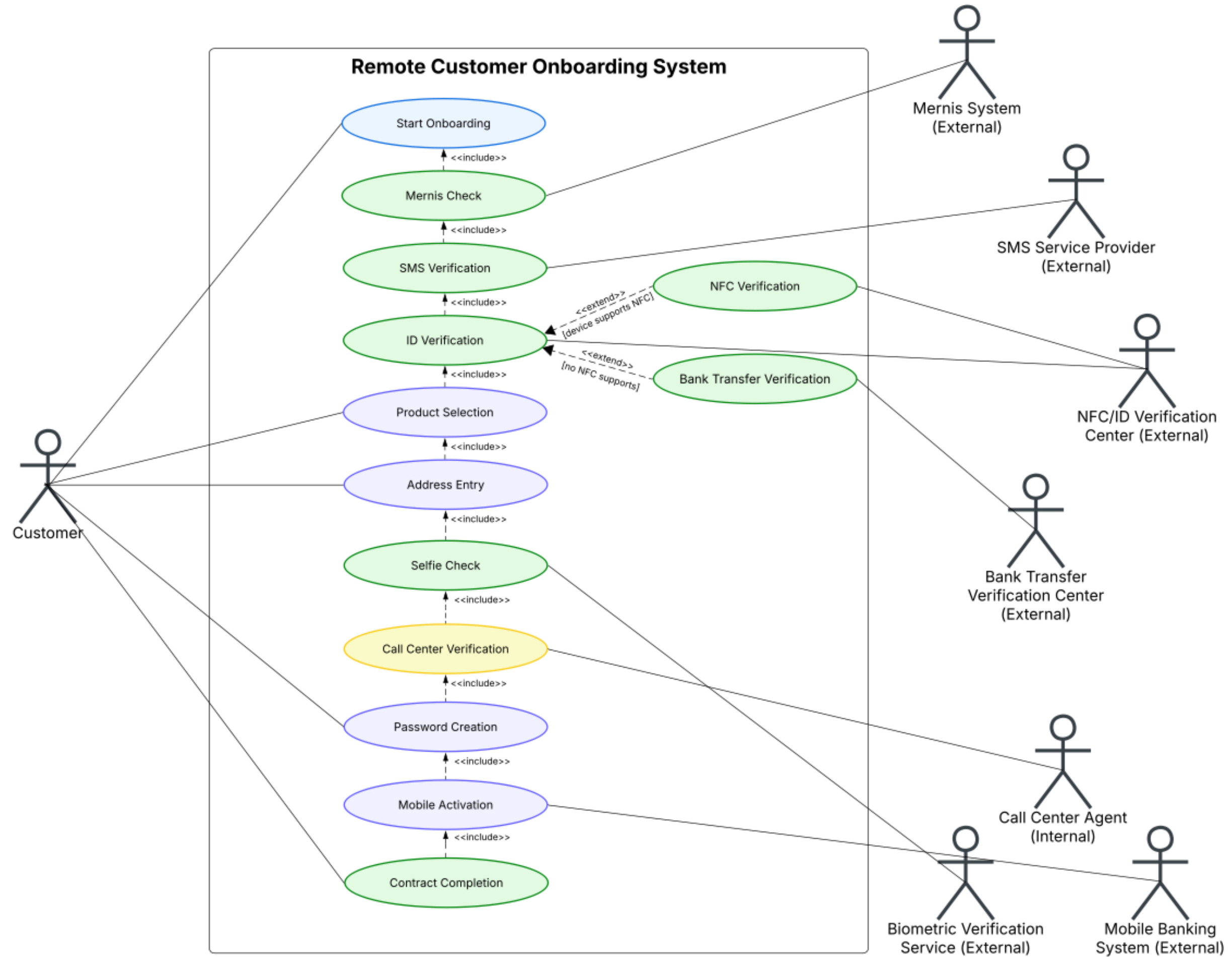
Event Table

#	Event Name	Trigger	Source	Activity	Response	Destination
1	Start Onboarding	User starts application	Customer	User fills initial application form	System initiates onboarding process	System
2	Mernis Check	Identity info entered	System	System sends identity data to Mernis service	Validate identity via Mernis	External System
3	SMS Verification	Phone number entered	System	System generates OTP and waits for input	Send OTP and verify	Customer
4	ID Verification	ID uploaded/scanned	Customer	User uploads/scans ID document	Validate ID information	System
5	Product Selection	User selects product	Customer	User browses and selects product	Save selected product	Database
6	Address Entry	Address entered	Customer	User enters address details	Store address info	Database
7	NFC Verification	NFC scan started	Customer	System reads NFC chip from ID	Read chip data	System
8	Selfie Check	Selfie captured	Customer	User captures selfie	Face verification	System
9	Call Center Verification	Video call started	System	Agent performs video identity check	Agent verifies identity	Call Center
10	Password Creation	User defines password	Customer	User creates password	Save credentials	Database
11	Mobile Activation	Activation triggered	System	System provisions mobile banking account	Activate mobile banking	System
12	Contract Completion	Process completed	System	System generates and finalizes contract	Send contract to customer	Customer

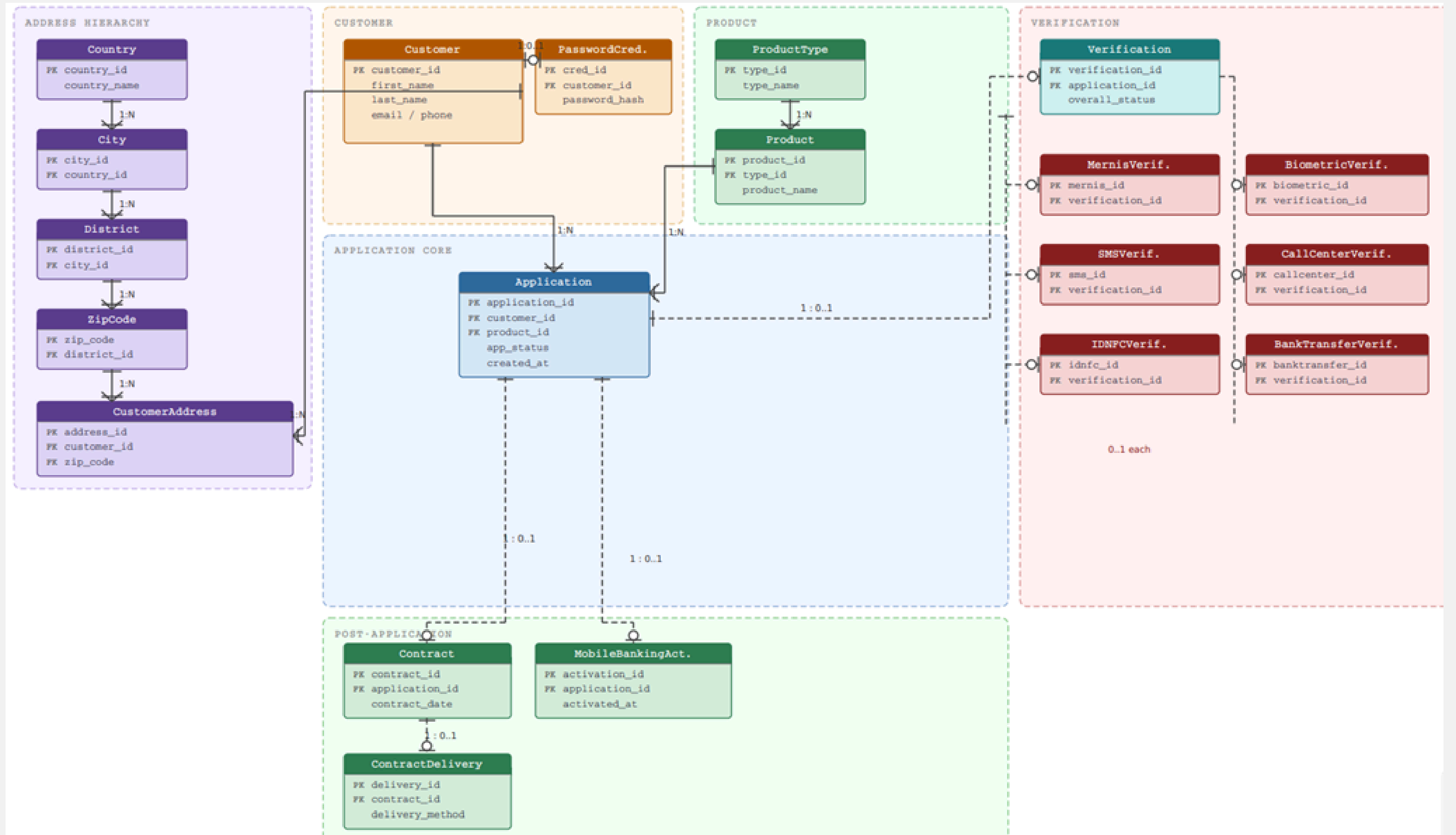
Context Level Diagram



Use Case Diagram



ER Diagram



Sequence Diagram



Scenario: Successful Verification Process

In this case, a customer uses the Remote Customer Onboarding System to successfully finish the verification procedure. When the customer starts the onboarding process, the system commences identity verification, which starts the verification process.

To confirm the customer's identity, the Verification System first transmits their identity details to the external Mernis System. After verifying the validity of the identity data, the Mernis System sends the outcome back to the Verification System.

The system then uses the SMS Service Provider to send a one-time password (OTP) to the customer's phone number. After receiving the SMS code, the client inputs it into the system. The SMS Service Provider receives the code from the Verification System and verifies its validity.

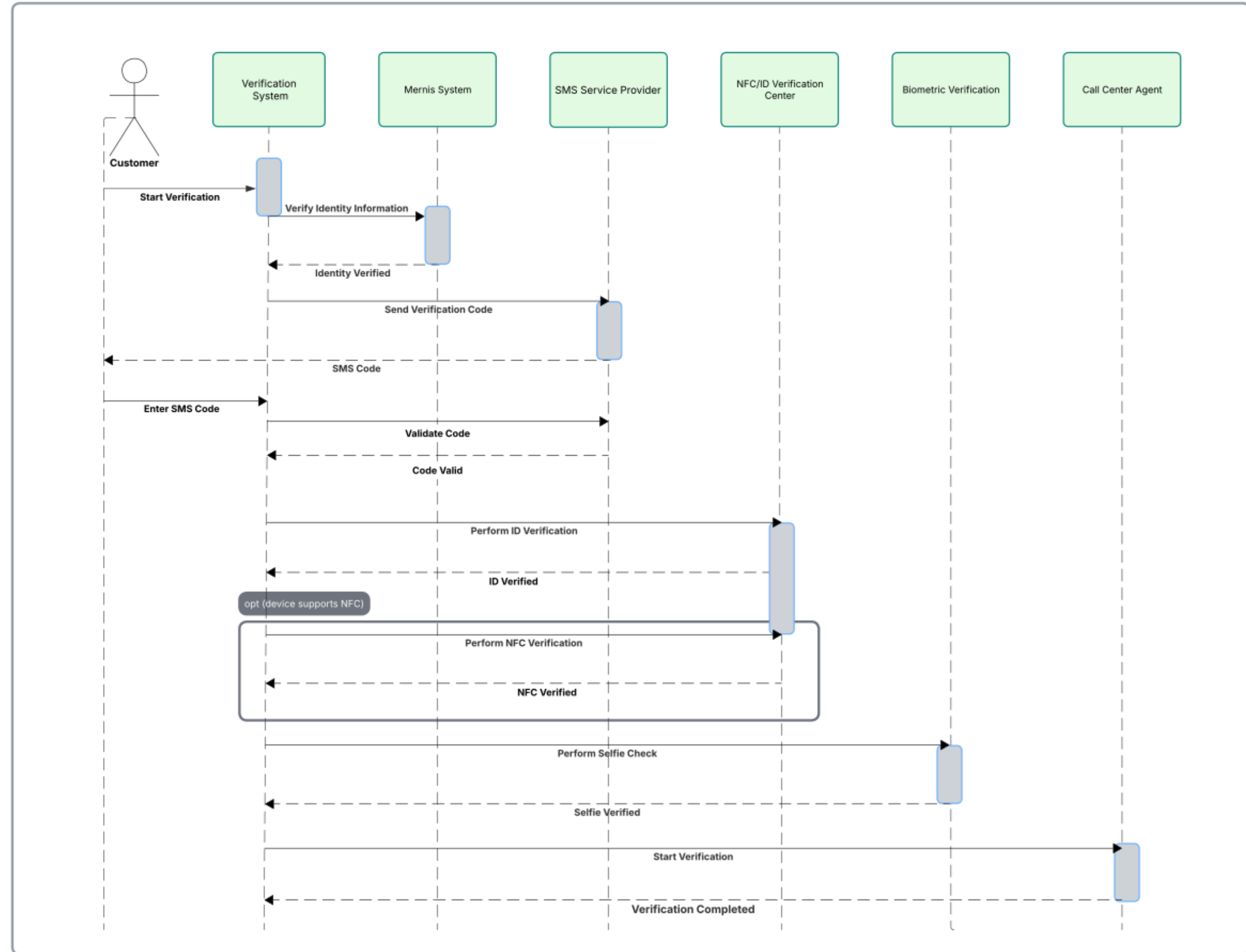
The NFC/ID Verification Center is then used by the system to verify identities. The system further completes NFC verification via the NFC/ID Verification Center as an optional step because the customer's device supports NFC. The NFC verification has been successfully finished.

The system uses the Biometric Verification Service to take a selfie after the ID is verified. The Biometric Verification Service receives the customer's selfie, examines it, and verifies that the liveness check was successful.

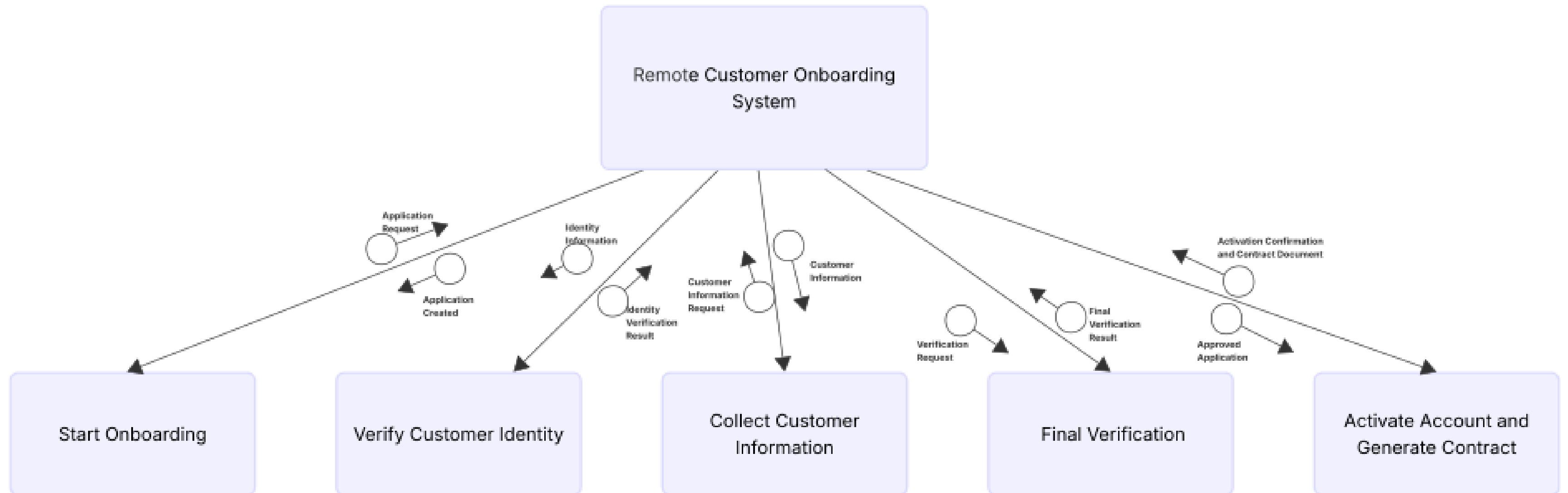
Call center verification is a required step in this system. Consequently, the Verification System notifies the Call Center Agent to conduct a manual verification once the selfie check is over. After reviewing the customer's data, the call center agent verifies that the verification process was successful. The customer is then informed by the Verification System that the verification process is complete and that all verification processes have been successfully performed.

Sequence Diagram

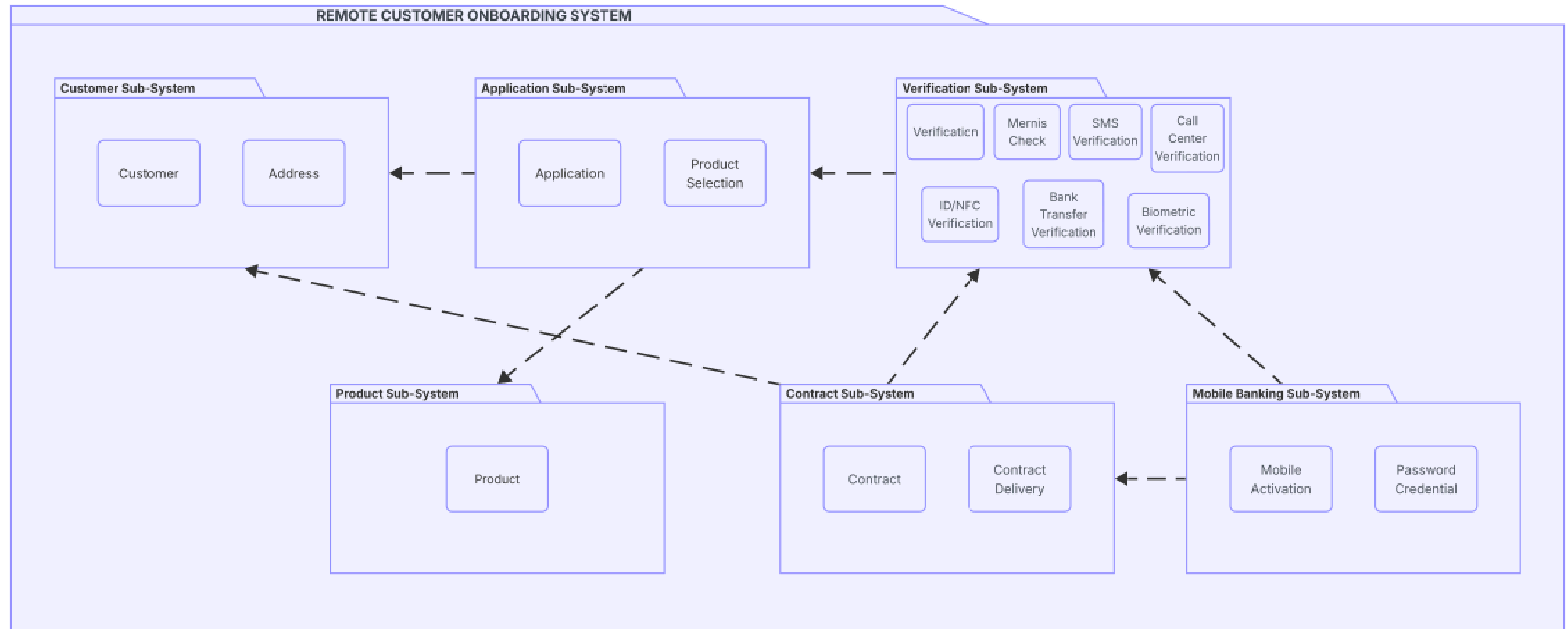
Remote Customer Onboarding System- Verification



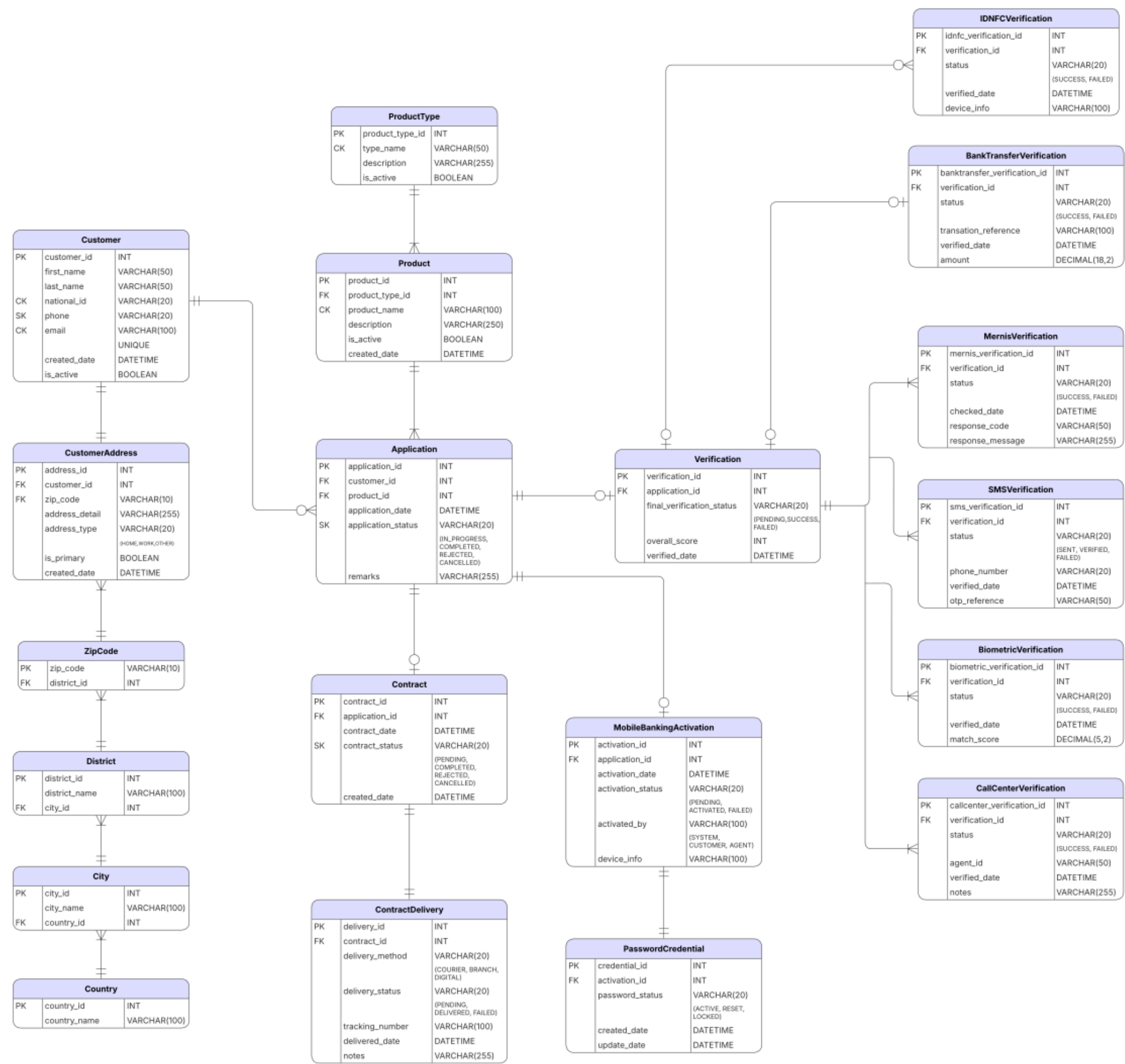
Top Structure Chart



Package Diagram with Classes



Database Design



Outputs

The system produces 7 main outputs:

1. Application Confirmation Notice — Confirms the application has been received
2. OTP / SMS Verification Result — Indicates whether SMS verification succeeded or failed
3. Identity Verification Summary — Consolidated result of Mernis, NFC/ID, and biometric checks
4. Contract Document — Final contract delivered to the customer
5. Mobile Banking Activation Confirmation — Confirms mobile banking activation
6. Onboarding Report — Daily/weekly statistics for bank managers
7. Rejection / Failure Notice — Informs the customer when onboarding cannot be completed

Three outputs designed in detail:

- Application Confirmation Notice: Includes Application ID, customer name, selected product, status, and next step. Sent only to the registered email/phone.
- Identity Verification Summary: Lists the status, timestamp, and source of each verification step. Stored in encrypted form.
- Onboarding Report (Weekly): Shows total applications, success rate (86.3%), failure counts per step, and average duration (13 min 42 sec). Restricted to authorized managers only.

Inputs

The system collects 7 main inputs:

1. Application Registration Input — Personal information submission
2. OTP Code Input — Phone verification code
3. ID Document Upload / NFC Read Input — Identity document or NFC data
4. Selfie / Biometric Capture Input — Live photo for face matching
5. Address Information Input — Residential and mailing address
6. Product Selection Input — Selected banking product
7. Password Definition Input — Mobile banking password

Three inputs designed in detail:

- Application Registration: First name, last name, National ID (11 digits, Mernis-validated), date of birth (18+), email, and phone number. Uniqueness check and rate limiting are applied.
- OTP Code Input: 6-digit numeric code, valid for 180 seconds, with a maximum of 3 retry attempts.
- Selfie / Biometric Capture: Min 480p, max 5 MB, single face detection, liveness check, and face match score ≥ 0.85 .

All inputs use a Simple Sequence Code, and supported devices include mobile, tablet, and desktop.

Menu Hierarchy

The system menu is divided into 7 main sections, serving both customers and bank staff:

1. Customer Portal — Start new application, resume application, status tracking, contract download
2. Verification Subsystem — Mernis, SMS/OTP, ID/NFC, Biometric, and Call Center verifications
3. Product & Application Management — Browse products, select product, application history
4. Address & Profile Management — Add/edit address, contact info, security settings
5. Contract & Activation — View contract, delivery method, mobile banking activation
6. Bank Officer / Admin Panel — Application review, verification queue, reports, failed applications
7. Support & Help — FAQ, live chat, call center contact, language settings

Başlangıç Zamanı 

3/23/2026

4/9/2026

18 days

Number of Cases 48,222

Average Case Duration 0

14 minutes

Trend Analysis

SDN 7 QDN

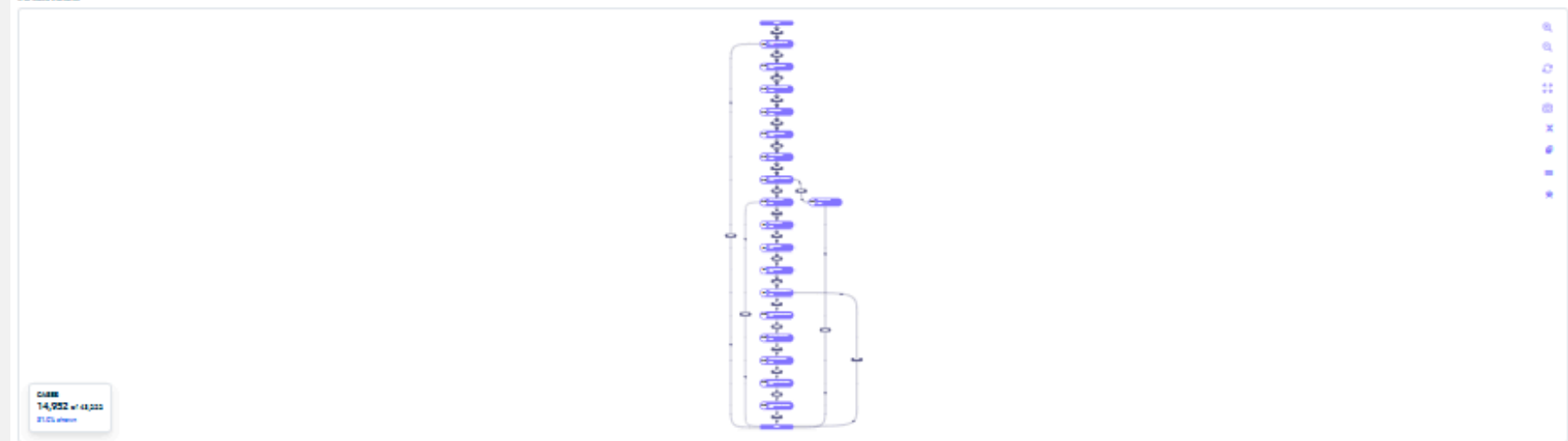
13.10%  -5.89%

0

Aktifite Telerantion İqarın Vokolar 41%

SÜREÇ HARİTASI

Word Count Worksheet



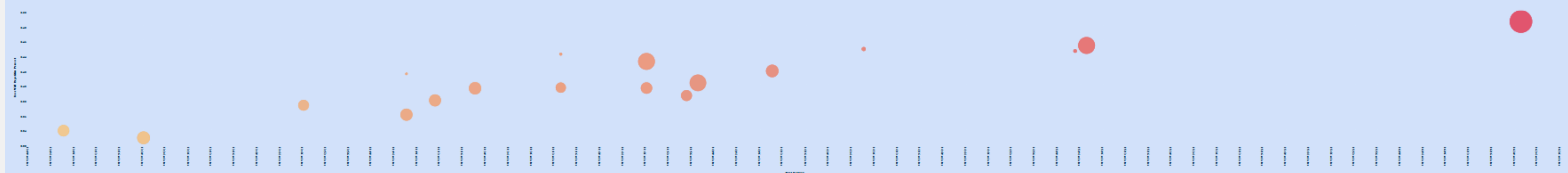
Aktivite Teleskopiinin Suresse Etilisi

Report

Activity	Number of Cases With Repetition	Average Repetitions Per Case	Average Added Time Per Repetition	Total Added Time Per Case	Total Added Time	15
BIB Co Volume	22881	0.07	11 Hours	1.23 Days	22 Years	
VIB Co Volume	10582	0.08	8 Hours	28 Hours	27 Years	
Vernon House Repetition	10285	0.46	18 Hours	1.87 Days	45 Years	
Bart's Dental	10210	1.06	11 Hours	27 Hours	27 Years	
VPS Co Volume	10082	0.06	10 Hours	27 Hours	26 Years	
GEO (442) Old Green Bay Repetition (144)	21200	0.6	8 Hours	10 Hours	20 Years	
GEO Co Vile	2007	1.00	10 Hours	1.00 Days	1 Years	
Bart's (144) Dental	2048	1.18	10 Hours	1.0 Hours	0 Years	
Hemlock (144) 2048 144	1170	1.18	10 Hours	10 Hours	0 Years	
Barnes' (144) 2048	877	0.86	10 Hours	0.86 Hours	0 Years	

Titled *Boys: 23*

Çizim Süre ve Rework Analizleri



Variant DNA (top 100)

Contributor ☐ Show Activity Names ☒

[illegible]

Test Case

Test Scenario: SMS OTP Verification

Test Case Code: SMS-TC-01

Preconditions:

- Customer has started a new onboarding application
- Mernis identity check has been successfully completed
- An OTP code has been sent to the customer's phone and has not yet expired (within 180 seconds)

Steps:

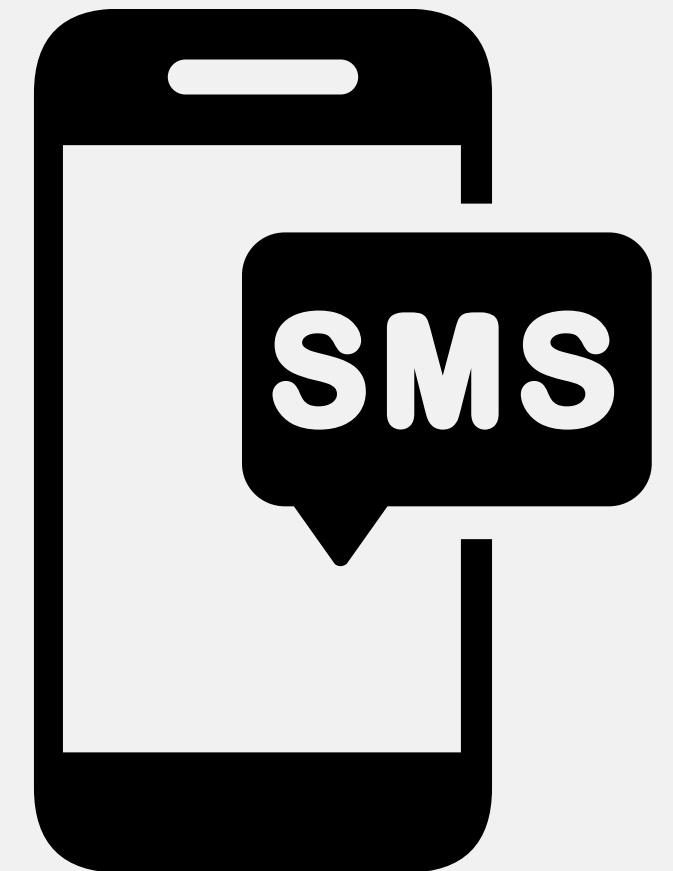
1. The system displays the OTP input field
2. The customer enters a 6-digit code
3. The customer clicks the "Verify" button

Test Data:

- System-generated OTP: 482915
- Customer-entered OTP: 123456

Expected Result: The system displays the error message "The OTP code you entered is incorrect. Please try again."
The verification status is set to "Failed," and the customer is prompted to retry (up to 3 attempts).

Purpose: To verify that the system correctly handles invalid OTP entries and enforces the security control for phone verification.



Do you have any questions?

Thank you for your patience!

